with climate change occurring, weather is becoming more and more unpredictable. It's a tug of war between weather and technology, and mother nature still has the upper hand.

That said, the one thing that is needed in order to predict the weather more accurately is more sensors, and of course, computing power to crunch the numbers in near-real-time, in order to be useful. With a country as big as India, and with the wild variation of weather across the landmass, obviously we need all the number crunching capacity we can get our hands on. Up until last year, India had the eighth-most powerful weather prediction system, with a number crunching capacity of 1 petaflop. This changed drastically when the Indian institute of Tropical Meteorology (IITM), Pune, unveiled a new supercomputer called Pratyush, dedicated to meteorology, with a massive 4 petaflops of capacity. Simultaneously the National Centre for Medium Range Weather Forecast (NCMRWF) in Noida got an upgrade to 2.8 petaflops capacity, which brings India's total to 6.8 petaflops. That's a huge step up from 1 petaflop, and puts India into fourth place in the world when it comes to weather computing power. Only UK, Japan and the US have more weather computing capacity than India now.

Because of the sheer computing power, India will now be able to much more accurately predict the monsoon, which is the season that the average farmer's livelihood depends on. We will also be able to forsee massive spikes in rainfall, and get more information out sooner, so as to help save lives with warnings, and prevent flooding.

More importantly, for farmers, this means better accuracy of information on medium and extended range weather forecasts, so that they can seed their crops on time for the monsoon. Not getting accurate information has been the biggest crib of farmers over the years.

Of course, while weather computing capacity puts us among the world's elite countries, numerous reports from Indian journalists have pointed out the biggest problem we still face – a lack of skilled software programmers who can help modify and adapt the existing computation models, and thus improve the forecasts. No matter how much computing power you throw at a problem... if the algorithm itself is not accurate, the results are going to be wrong. So what are you waiting for? If you're a skilled coder, contact IITM and try and get a job there to help out hundreds of millions of farmers across the country.

TYPES OF FORECASTS

- **Day forecast:** Also called 'nowcast', is the most accurate weather prediction, and is for the next 24 hours.
- **Short-range:** This is a forecast of the next 72 hours (3 days)
- **Medium-range:** Between 3 and 10 days
- **Extended range:** 10 to 30 days
- **Long range:** Seasonal, or yearly. Is also the least accurate and least relied upon of all forecasts.

SOIL

While the Indian subcontinent is considered to be some of the most fertile land on earth, it's also been farmed for millenia. With an ever increasing population, the farmlands need replenishing and obviously the soil needs to be protected and brought back to life. We are far from peak efficiency when it comes to managing the soil, and this is something we're going to need to do in order to feed the country. With an expected population increase to 1.7 billion by 2050, at at the same time a goal to get rid of the hunger that plagues almost 200 million people in India, we need to get farmers to manage the soil quality on their lands.

hile it seems odd to start with this, it's one way of dealing with the problem of depleted soil... Hydroponics is the science of growing plants without soil, and substituting soil with sand, gravel, or even water. The biggest advantage of hydroponics is that it can assist in vertical farming (farming in a skyscraper) which yields far more crops per given area of land. This makes it perfect for farming within city limits, and also allows smaller, otherwise arid plots of land to be used to continue farming.

However, coming to the soil itself, farmers have consistently struggled to find the perfect balance of nutrients and soil health in order to maximise their crop output. In fact, a lot of farmers have been overusing chemical fertilisers, which is believed to be bad in the long run. Then there's the lack of crop rotation, and of course the lack of funds to use complicated testing methods to know the status of their farm's soil. The space was crying out for a cost effective test for soil conditions, and a Karnataka-based startup called EasyKrishi might just have the solution. They recently showcased a prototype at the Kerala Startup Mission's Global Impact Challenge 2018 in Thiruvananthapuram. Their kit (which is also app supported), can tell a farmer the pH and NPK (Nitrogen, Phosphorus and Potassium) levels of the soil in minutes.


Image: Triton Foodworks

Triton Foodworks is a startup working on hydroponics in Delhi

It's not just individual farmers that need to know about the condition of the soil on their farms, it's also banks who want to know the conditions in much larger areas, encompassing thousands of farmers. Banks, are, after all, for profit insti-